AutoGrid Sentinel

Smart Power Source Monitoring and Automatic Transfer Switch

1. Abstract

AutoGrid Sentinel is an Arduino based Automatic Transfer Switch (ATS). The system continuously monitors the main power supply. If the main supply fails, it automatically disconnects the main source and connects the generator to the load. When the main supply returns, the system switches back automatically. A 20x4 LCD displays the current power source status.

2. Objectives

• Ensure uninterrupted power supply.
• Automatically switch between Main and Generator.
• Display system status on LCD.
• Build and test the design in Proteus.

3. Components Used

Arduino UNO
20x4 LCD
Step-down Transformer
Diode
10uF Capacitor
40kΩ and 10kΩ Resistors (Voltage Divider)
Two 5V Relays
LED Indicator
Load (Lamp)
Proteus 8 Professional

4. Working Principle

The transformer reduces the AC voltage. The diode converts AC to DC and the capacitor smooths the voltage. A voltage divider reduces the voltage to a safe level for the Arduino analog pin. The Arduino continuously checks the main supply. If the voltage is present, the main relay remains ON and the generator relay remains OFF. If the main supply fails, the Arduino turns OFF the main relay and turns ON the generator relay. The LCD displays the active source.

5. Advantages

Automatic operation.
Reduced power interruption.
Simple and low cost.
Easy to expand with inverter or IoT.

6. Future Scope

Add inverter support, IoT monitoring, GSM alerts, voltage and current monitoring, fault logging, and mobile application support.

7. Software Used

Arduino IDE and Proteus 8 Professional.

8. Conclusion

The project successfully demonstrates automatic power source transfer between the main supply and generator using Arduino and relays.